Kalman Filter, Localization and Mapping Continued

Robots in Context

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Agenda & Content

- Programming Exercise 6.2, PRM / RRT
  - Frederik
  - Marius
- ➢ Exercise 7, KF due: 17/11
  - ➢
- Summary from last lecture
- Localization using range and bearing
  - Data Association
- Simultaneous Localization and Mapping
- Probabilistic Localization without a Kalman Filter
- Brief intro to Mapping



		PE1	PE2	PE3	PE4	PE5	PE6.1	PE6.2	PE7
Alexander Kiraly Hansen	1					1			
Amr Almaradni	0								
Anders Christiansen	0								
Asmus Snede Rise	0								
August Maximillian Rosford Tranberg	1				1				
Dominik Felten	1			1					
Elias Ottar Alstrup	0								
Emil Klitgaard	1						1		
Frederik Svend Vester	1							1	
Jacob Grud Agerbæk Madsen	1					1			
Jakob Tietgen Vedel Eriksen	1		1						
Jeppe Rønnow	0								
Jonathan Balder Dietrich	0								
Karl Oskar Ellgaard Soneff	1	1							
Kristoffer Nielsen	0								
Lukas Hjeronimus Sørensen	1				1				
Mads Bendorff Thomasen	1		1						
Mads Riis Olsen	0								
Magnus Meldgård	1	1							
Marius Simmelsgaard Krintel	1							1	
Povl Bonde Christiansen	1			1					
Rune Kildahl Frederiksen	0								



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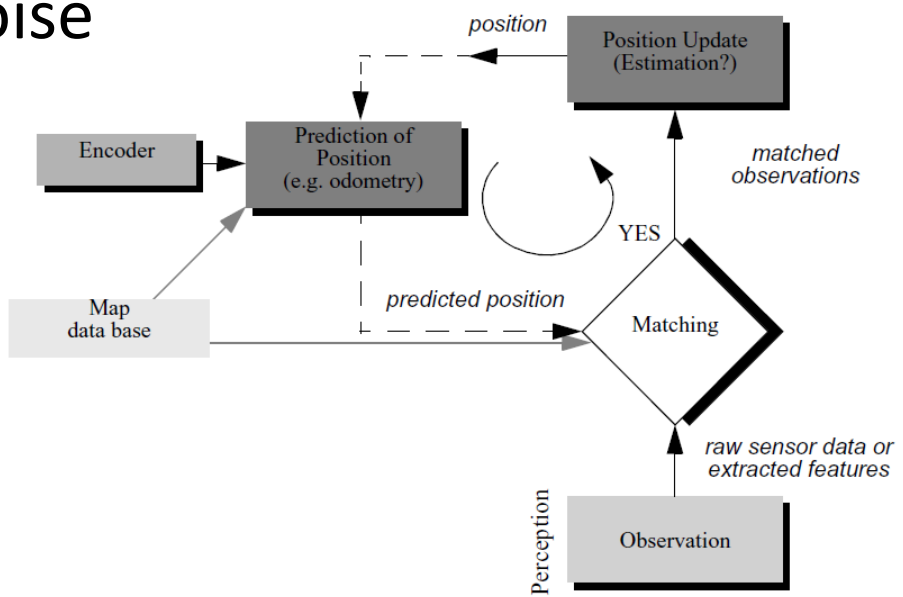
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Kalman Filter

- Assumptions
  - Unimodal
  - Zero mean Gaussian noise
  - Linear system
- Requirements
  - Define the state
  - System model
  - Observation Model

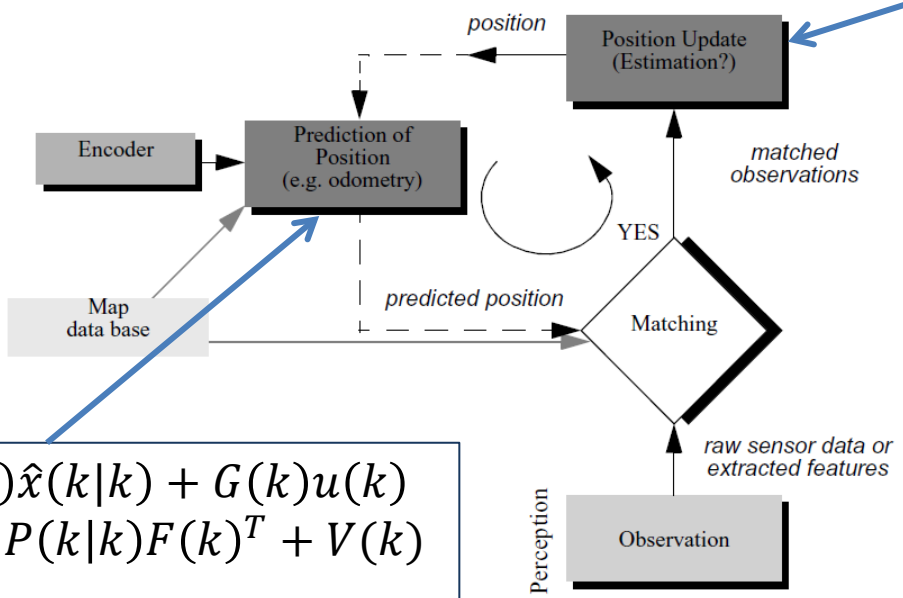


Localization using Kalman Filter

$$\hat{x}(k+1|k+1) = \hat{x}(k+1|k) + R\,v$$
$$P(k+1|k+1) = P(k+1|k) + RHP(k+1|k)$$

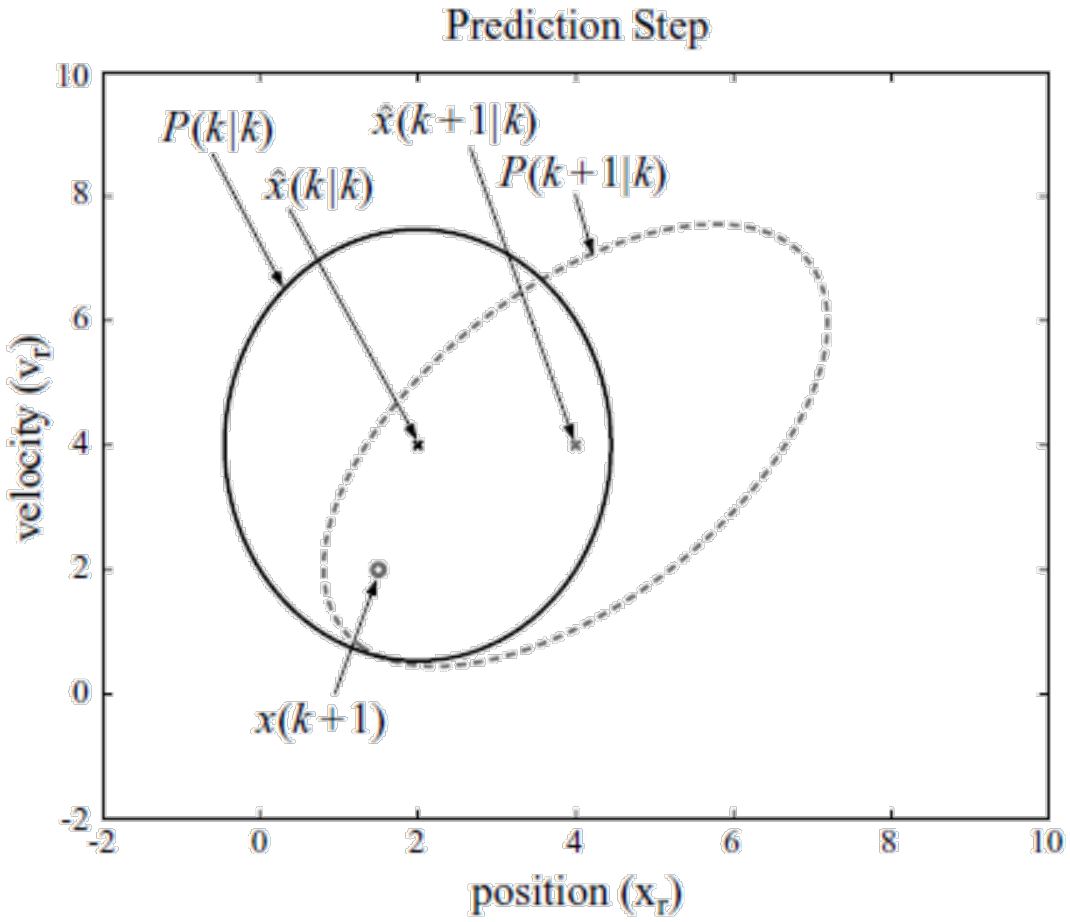
where

$$v = y(k+1) - H\hat{x}(k+1|k)$$
$$S = HP(k+1|k)H^T + W(k+1)$$
$$R = P(k+1|k)H^T S^{-1}$$

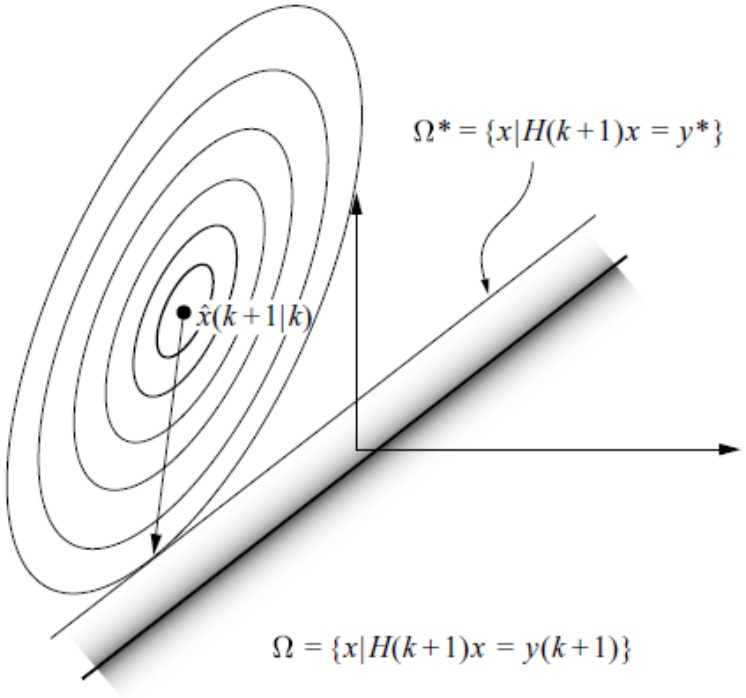


$$\hat{x}(k+1|k) = F(k)\hat{x}(k|k) + G(k)u(k)$$
$$P(k+1|k) = F(k)P(k|k)F(k)^T + V(k)$$

Kalman Filter – Prediction



Kalman Filter – Update



Extended Kalman Filter

- The ordinary Kalman Filter requires that the system and observation models are linear.
- In the Extended Kalman Filter, EKF, the system is given by

$$x(x+1) = f(x(k), u(k), k) + v(k)$$
$$y(k) = h(x(k), k) + w(k)$$

where *f* and *h* are continuously differentiable.



Extended Kalman Filter

- To run the *predict* and *update* step of the Kalman filter the Extended Kalman Filter makes use of the partial derivatives of *f* and *h*

- $$F(k) = \frac{\partial f}{\partial x}\Big|_{x=\hat{x}(k|k)} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

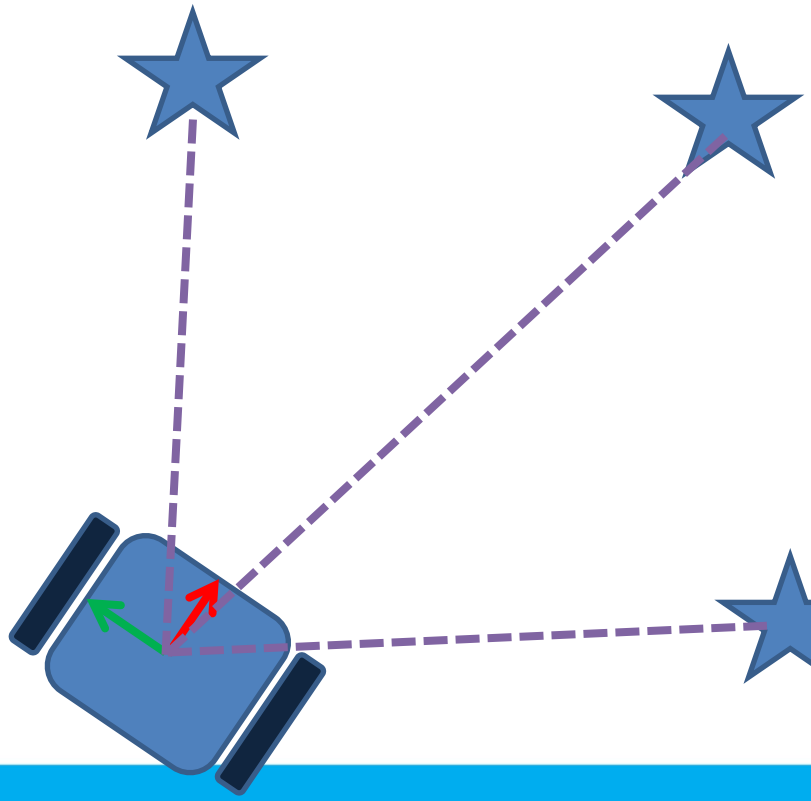
- $$H(k) = \frac{\partial h}{\partial x}\Big|_{x=\hat{x}(k+1|k)} = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_p}{\partial x_1} & \cdots & \frac{\partial h_p}{\partial x_n} \end{bmatrix}$$

- Whereever we cannot directly use *f* and *h*, we use above linearizations instead.



Localization using Range and Bearing

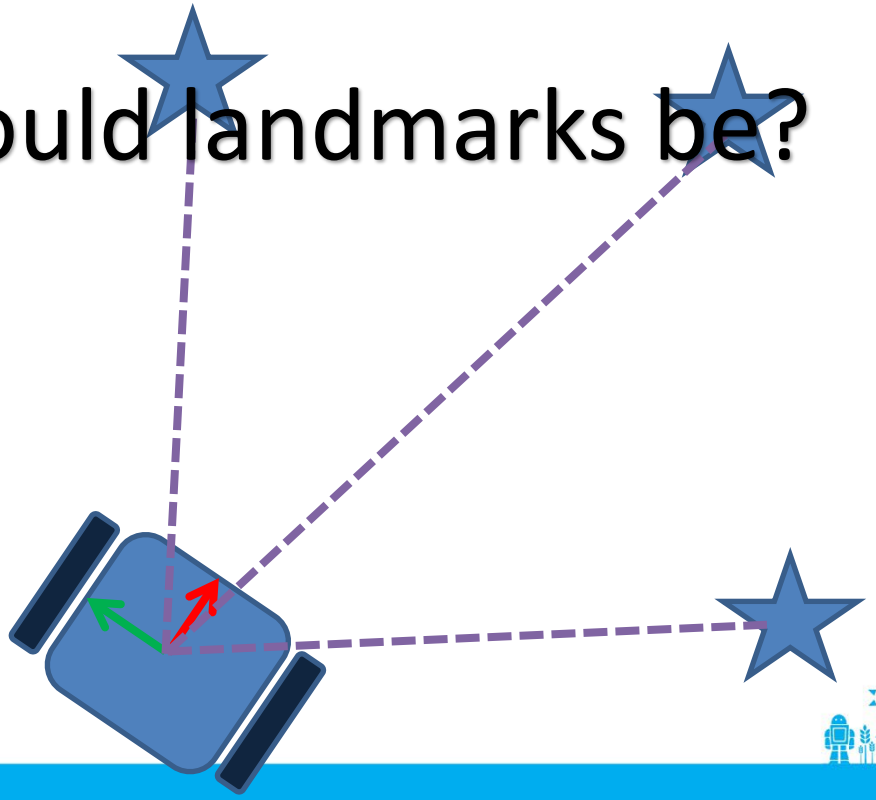
We can identify ”landmarks” and measure the range and direction relative to the robot.



Localization using Range and Bearing

We can identify ”landmarks” and measure the range and direction relative to the robot.

What could landmarks be?



The Robot

We will assume a robot with input

$$u(k) = \begin{bmatrix} \alpha(k) \\ \omega(k) \end{bmatrix}$$

where $\alpha(k)$ is the linear velocity and $\omega(k)$ is the angular velocity.

State?



The Robot

We will assume a robot with input

$$u(k) = \begin{bmatrix} \alpha(k) \\ \omega(k) \end{bmatrix}$$

where $\alpha(k)$ is the linear velocity and $\omega(k)$ is the angular velocity.

State?

$$x(k) = \begin{bmatrix} x_r(k) \\ y_r(k) \\ \theta_r(k) \end{bmatrix}$$



System/Proces Model

What is the System Model?

$$x(k+1) = f(x(k), u(k), k) + v(k)$$



System/Proces Model

What is the System Model?

$$x(k+1) = f(x(k), u(k), k) + v(k)$$

$$= \begin{bmatrix} \cos\left(\theta_r(k)\right)\alpha(k) + x_r(k) \\ \sin\left(\theta_r(k)\right)\alpha(k) + y_r(k) \\ \omega(k) + \theta_r(k) \end{bmatrix} + v(k)$$



Observation/Sensor Model

General Form

$$y(k) = h(x(k), u(k)) + w(k)$$

Given p(k) observations at time k

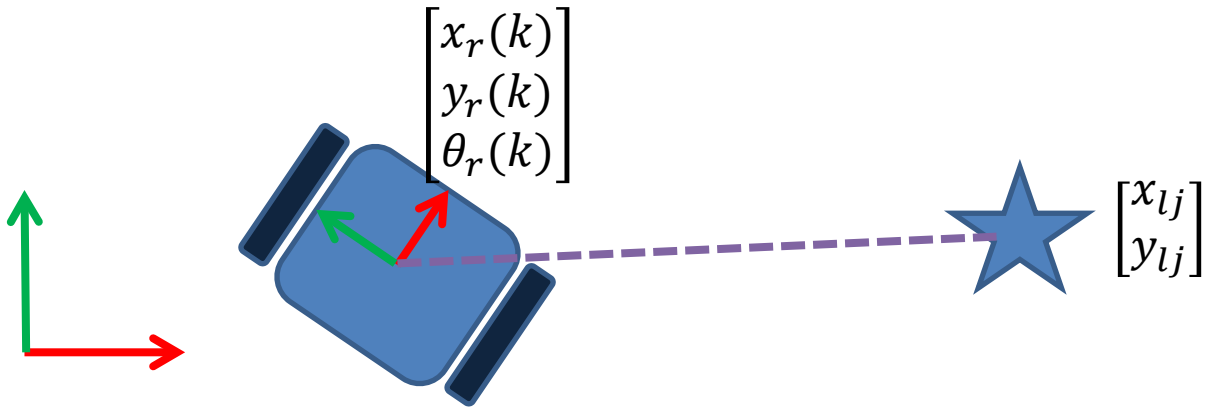
$$y(k) = \begin{bmatrix} h_1(x(k), a(1)) \\ h_2(x(k), a(2)) \\ \vdots \\ h_{p(k)}(x(k), a(p(k))) \end{bmatrix} + \begin{bmatrix} w_1(k) \\ w_2(k) \\ \vdots \\ w_{p(k)}(p(k)) \end{bmatrix}$$

where *a(i)* is a mapping between observation number *i* and landmark *i,* with *p(k)* observations available at time *k*.



Observation/Sensor Model

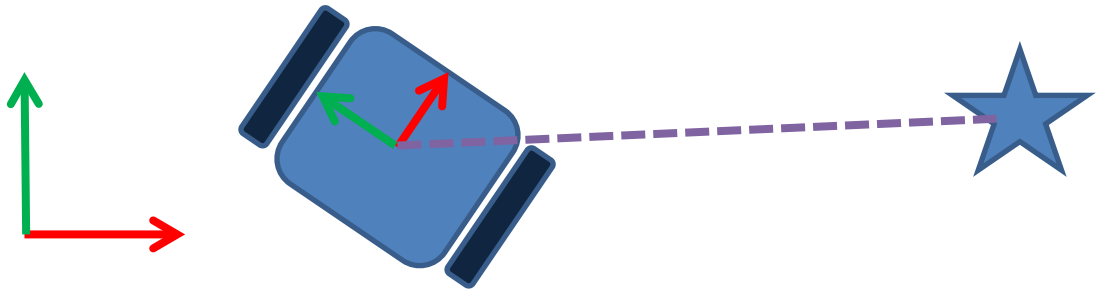
What is $h_i(x(k), j)$?



Observation/Sensor Model

Single Observation

$$h_i(x(k), j) = \begin{bmatrix} \sqrt{(x_r(k) - x_{lj})^2 + (y_r(k) - y_{lj})^2} \\ atan2\left(y_r(k) - y_{lj}, x_r(k) - x_{lj}\right) - \theta_k(k) \end{bmatrix}$$



Using the EKF

- To use the EKF we need to compute the partial derivatives of $f(x(k), u(k), k)$ and $h(x(k), k)$

$$f(x(k), u(k), k) = \begin{bmatrix} \cos\left(\theta_r(k)\right) \alpha(k) + x_r(k) \\ \sin\left(\theta_r(k)\right) \alpha(k) + y_r(k) \\ \omega(k) + \theta_r(k) \end{bmatrix}$$

$$F(x(k), u(k), k) = ?$$



Using the EKF

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$$F(x(k), u(k), k) = ?$$

Hint: $F_{i,j}(x(k), u(k), k) = \frac{\partial f_i\,(x(k), u(k), k)}{\partial x_j}$



Using the EKF

- To use the EKF we need to compute the partial derivatives of $f(x(k), u(k), k)$ and $h(x(k), k)$

$$f(x(k), u(k), k) = \begin{bmatrix} \cos\left(\theta_r(k)\right) \alpha(k) + x_r(k) \\ \sin\left(\theta_r(k)\right) \alpha(k) + y_r(k) \\ \omega(k) + \theta_r(k) \end{bmatrix}$$

$$F(x(k), u(k), k) = \begin{bmatrix} 1 & 0 & -\sin\left(\theta_r(k)\right)\alpha(k) \\ 0 & 1 & cos\left(\theta_r(k)\right)\alpha(k) \\ 0 & 0 & 1 \end{bmatrix}$$

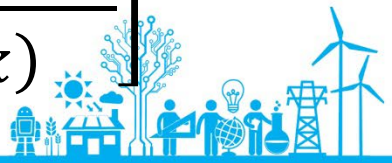


Using the EKF

- To use the EKF we need to compute the partial derivatives of $f(x(k), u(k), k)$ and $h(x(k), k)$

$$h_i(x(k), j) = \begin{bmatrix} \sqrt{(x_r\ (k) - x_{lj})^2 + (y_r\ (k) - y_{lj})^2} \\ atan2\left(y_r\ (k) - y_{lj}, x_r\ (k) - x_{lj}\right) - \theta_k(k) \end{bmatrix}$$

$$H_i(x(k), k) = \begin{bmatrix} \frac{\partial h_{i,1}(x(k), j)}{\partial x_r\ (k)} & \frac{\partial h_{i,1}(x(k), j)}{\partial y_r\ (k)} & \frac{\partial h_{i,1}(x(k), j)}{\partial \theta_r\ (k)} \\ \frac{\partial h_{i,2}(x(k), j)}{\partial x_r\ (k)} & \frac{\partial h_{i,2}(x(k), j)}{\partial y_r\ (k)} & \frac{\partial h_{i,2}(x(k), j)}{\partial \theta_r\ (k)} \end{bmatrix}$$

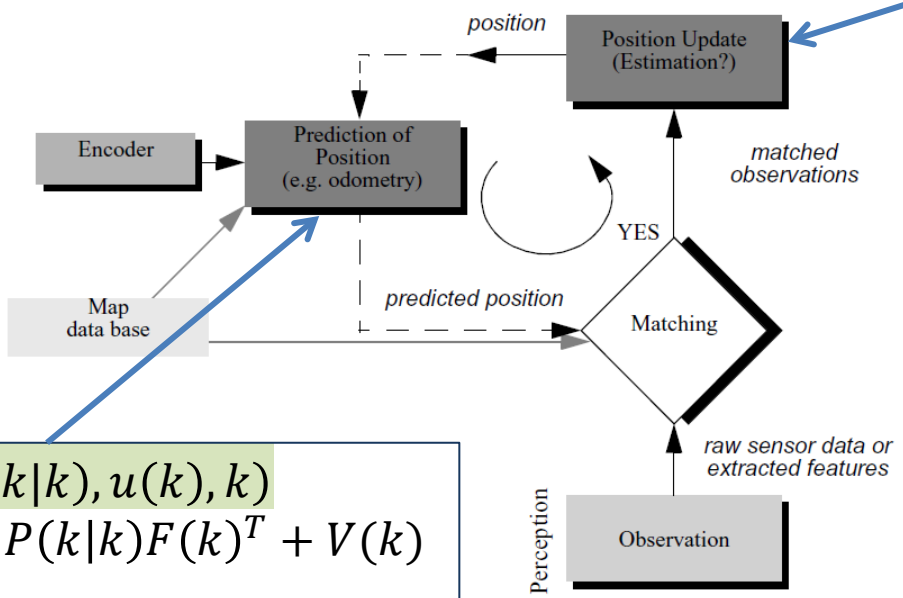


Localization using Kalman Filter

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$$P(k+1|k+1) = P(k+1|k) + RHP(k+1|k)$$

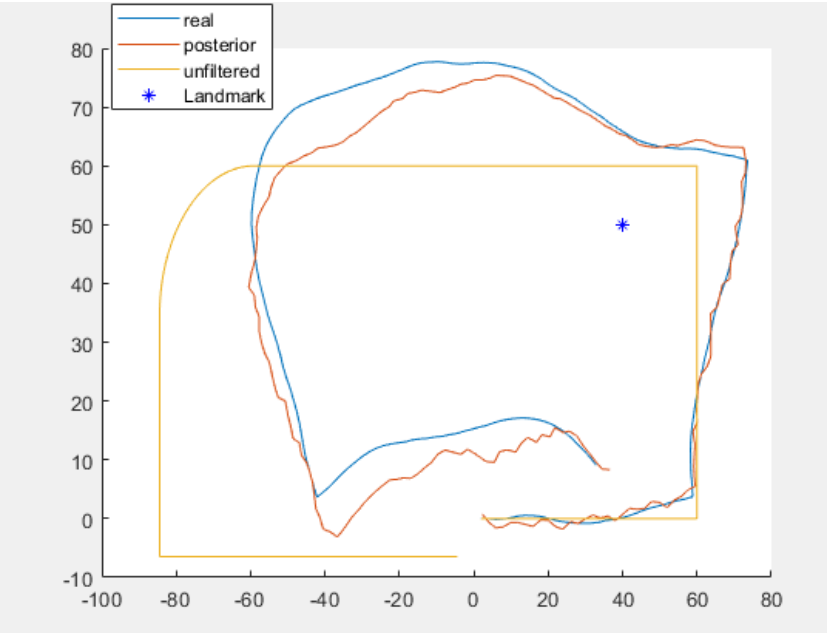
where

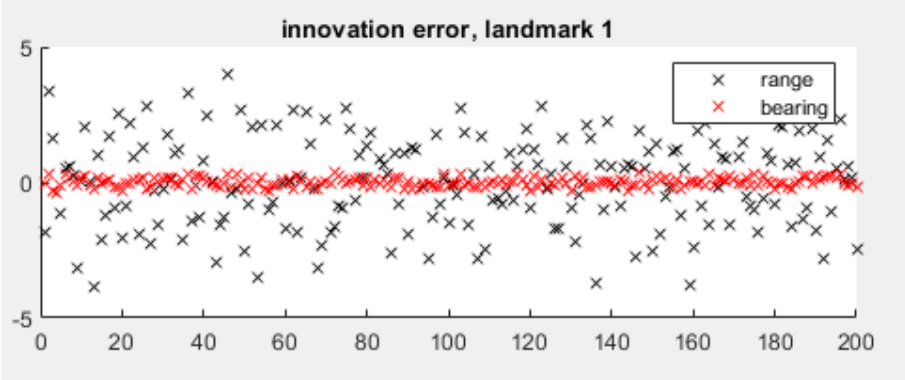
$$v = y(k+1) - h(\hat{x}(k+1|k))$$
$$S = HP(k+1|k)H^T + W(k+1)$$
$$R = P(k+1|k)H^T S^{-1}$$

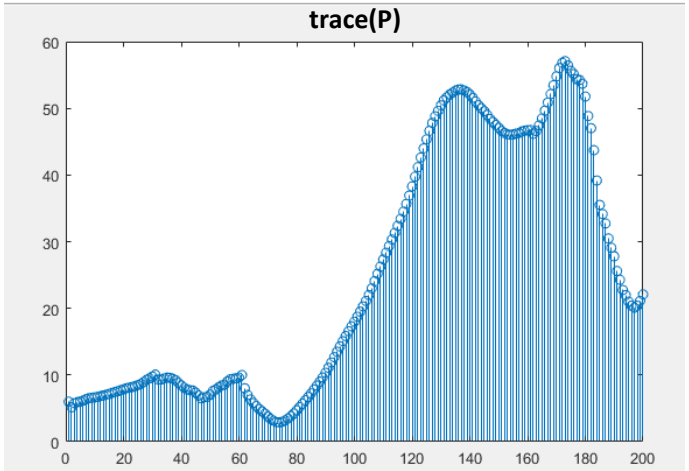


$$\hat{x}(k+1|k) = f(\hat{x}(k|k), u(k), k)$$
$$P(k+1|k) = F(k)P(k|k)F(k)^T + V(k)$$

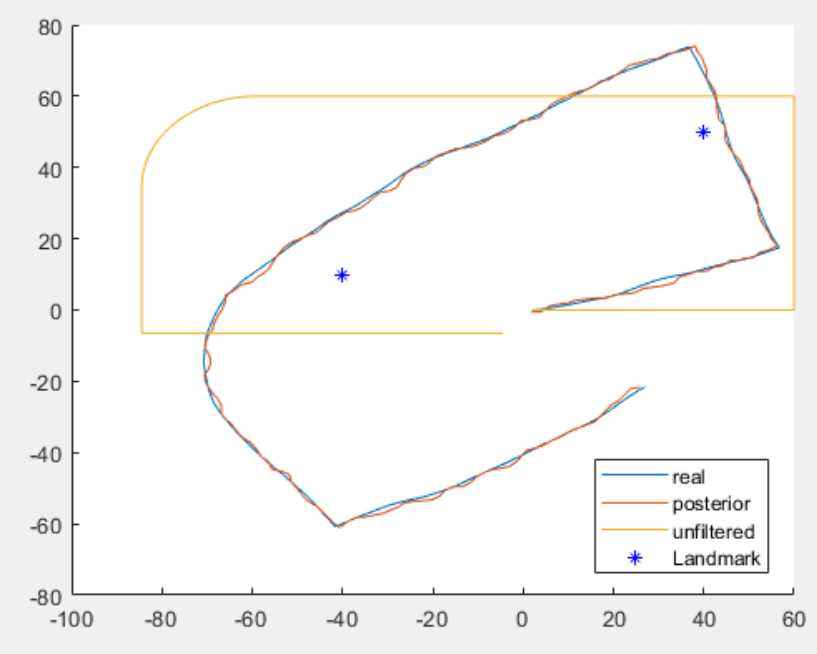
EKF with Range and Bearing, example

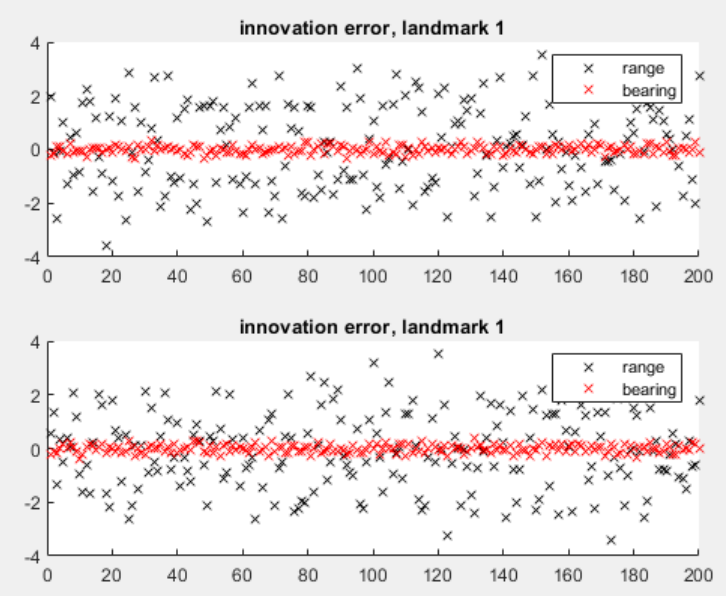


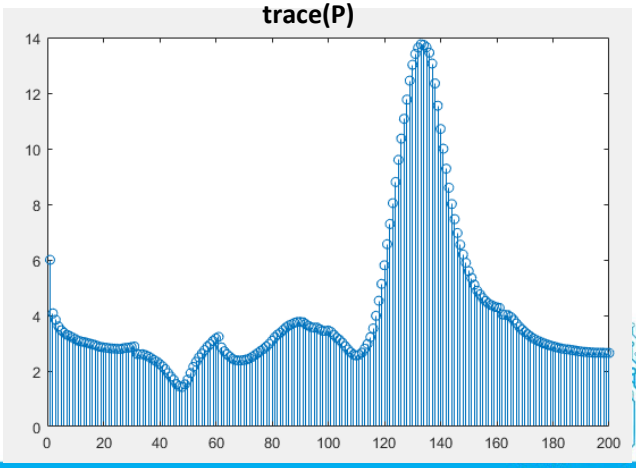




EKF with Range and Bearing, example

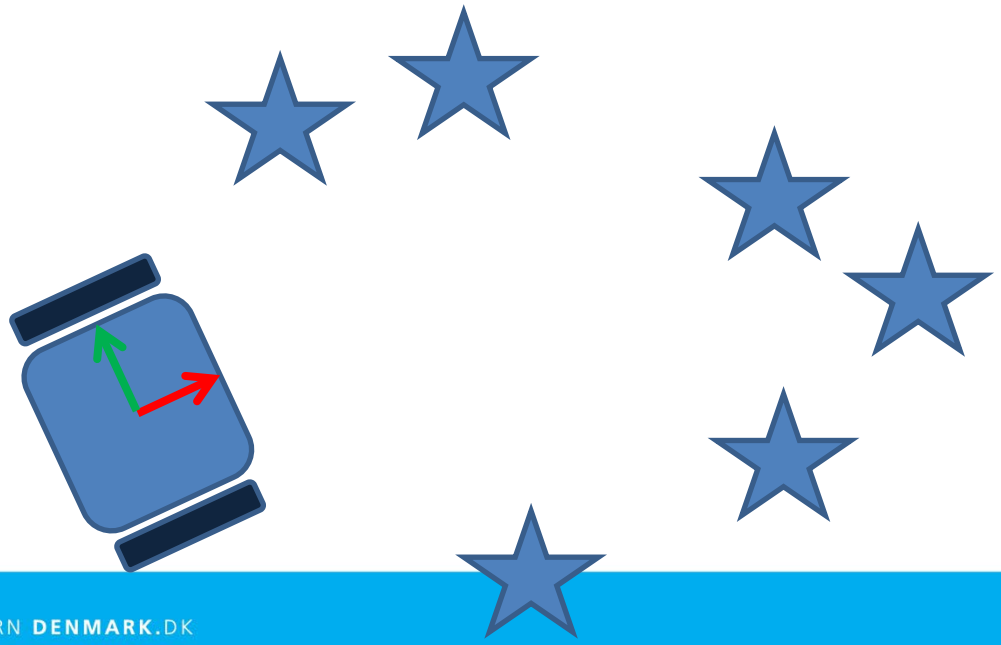






Data Association

- Previously we assumed to have a mapping between observations and landmarks.
- This problem is called Data Association



Data Association

- Given a measurement $y_i(k+1)$ and a predicted state $\hat{x}(k+1|k)$ we can for each landmark in the map compute the ”innovation”

$$v_{ij} = y_i(k+1) - h_i(\hat{x}(k+1|k), j)$$

- What does the innovation tell us?



Data Association

- Given a measurement $y_i(k+1)$ and a predicted state $\hat{x}(k+1|k)$ we can for each landmark in the map compute the ”innovation”

$$v_{ij} = y_i(k+1) - h_i(\hat{x}(k+1|k), j)$$

- What does the innovation tell us?
  - How ”close” a landmark is to where we expect it to be.



Data Association

$$v_{ij} = y_i(k+1) - h_i(\hat{x}(k+1|k), j)$$

Should we just take the landmark with smallest v_{ij}?



Data Association

$$v_{ij} = y_i(k+1) - h_i(\hat{x}(k+1|k), j)$$

Should we just take the landmark with smallest v_{ij}? No, we need to consider the uncertainties.

A uncertainty weighted innovation can be computed as $\chi^2 = v_{ij}(S_{ij})^{-1}{v_{ij}}^T$

Which is known as the **Mahalanobis** norm giving the distance in terms of ”number of standard deviations”



Data Association

$$\chi^2 = v_{ij}(S_{ij})^{-1}{v_{ij}}^T$$

To get the right uncertainty we need to consider both $P(k+1|k)$ and $W_i(k+1)$.

$$S_{ij} = H_i(k+1,j)P(k+1|k)H_i(k+1,j)^T + W_{ij}(k+1)$$

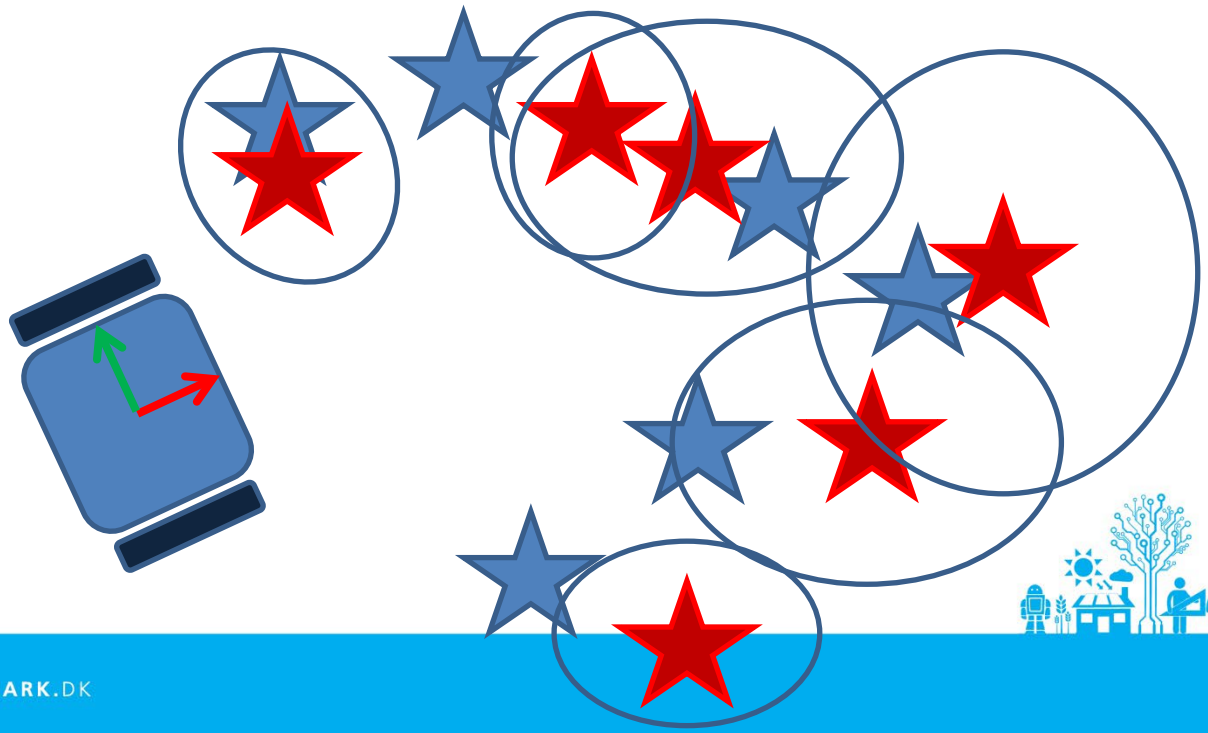
All of which are available from the Kalman Filter.



Data Association

Predicted Landmark

Detected Landmark with covariance



Localization with EKF One-Page Summary

- Set up system Model
- Set up observation Model
- Implement (Extended) Kalman Filter
- Implement Data Association
- Run the filter alternating between prediction and updating



Localization with EKF One-Page Summary

- Set up system Model
- Set up observation Model
- Implement (Extended) Kalman Filter
- Implement Data Association
- Run the filter alternating between prediction and updating

Do you always need the full prediction/update cycle?



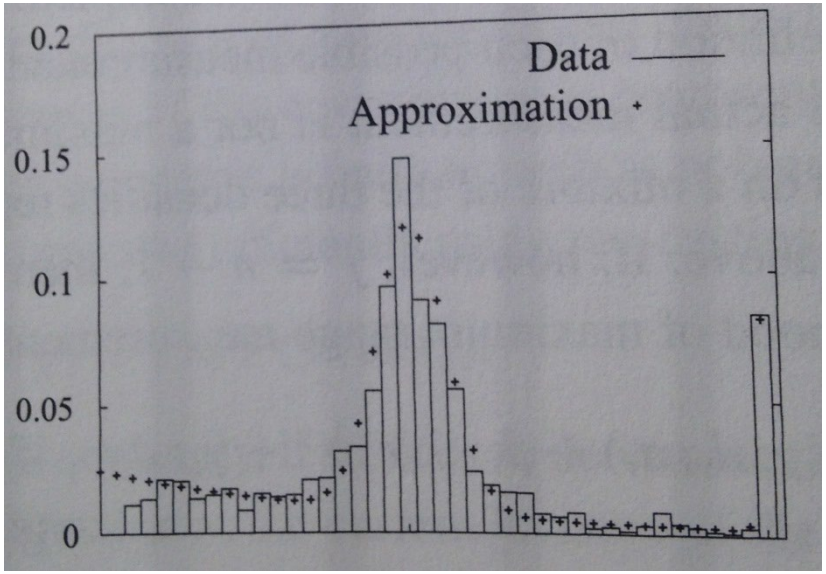
Mapping

- Creating a map of the environment, assuming we always know the location of the robot.
  - Easy right?



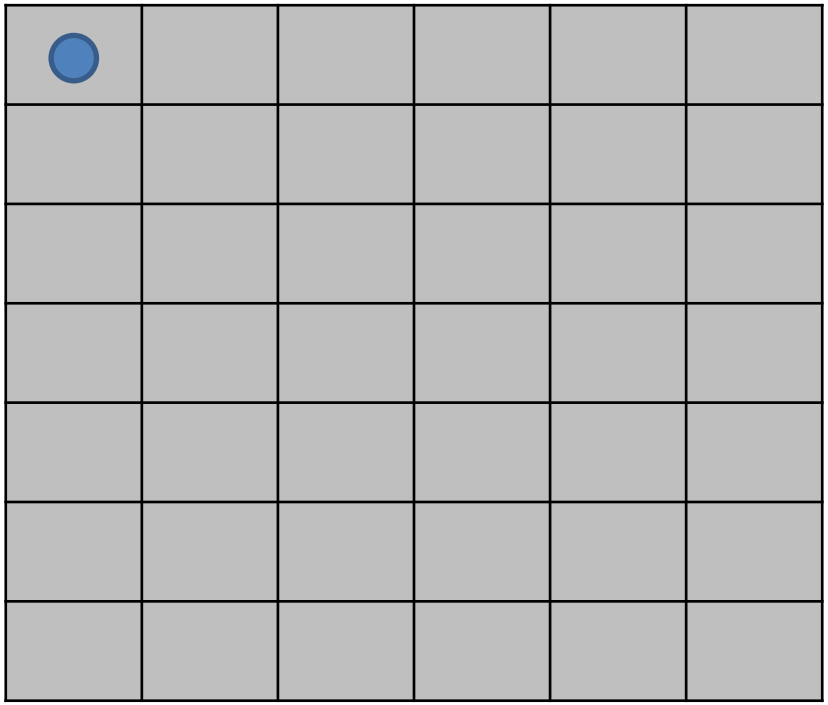
Mapping

- Mapping with known location of the robot
- Challenges
  - Sensor noise.
  - We only know what we actually ‘see’.

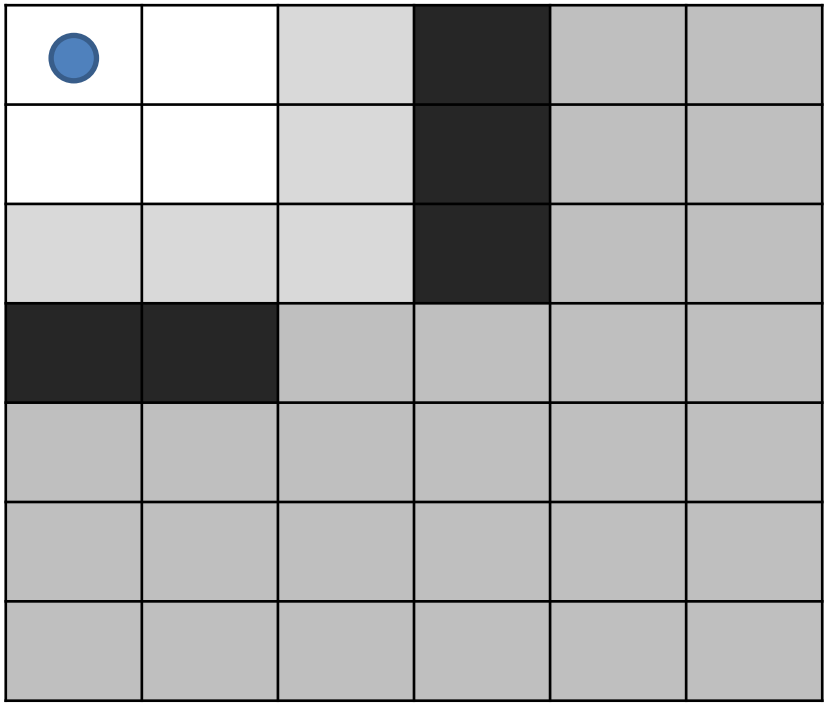


Mapping, Occupancy Grid

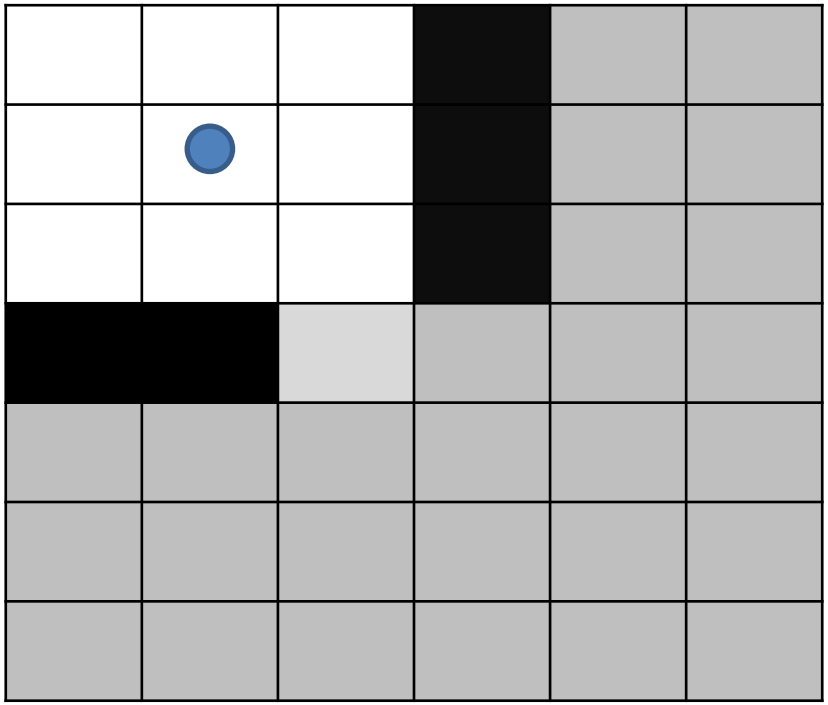
- We add up the knowledge we have



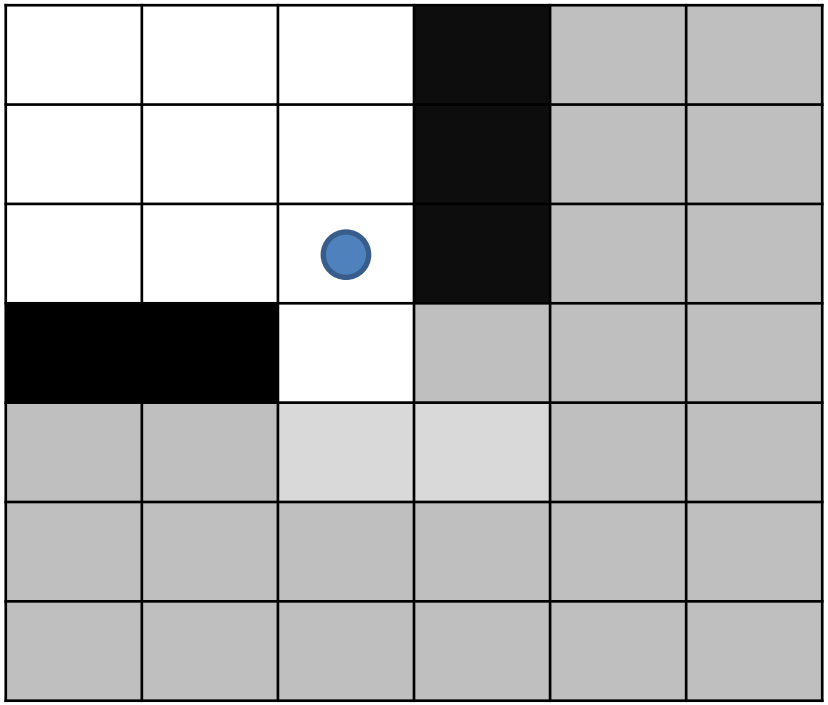
Mapping, Occupancy Grid



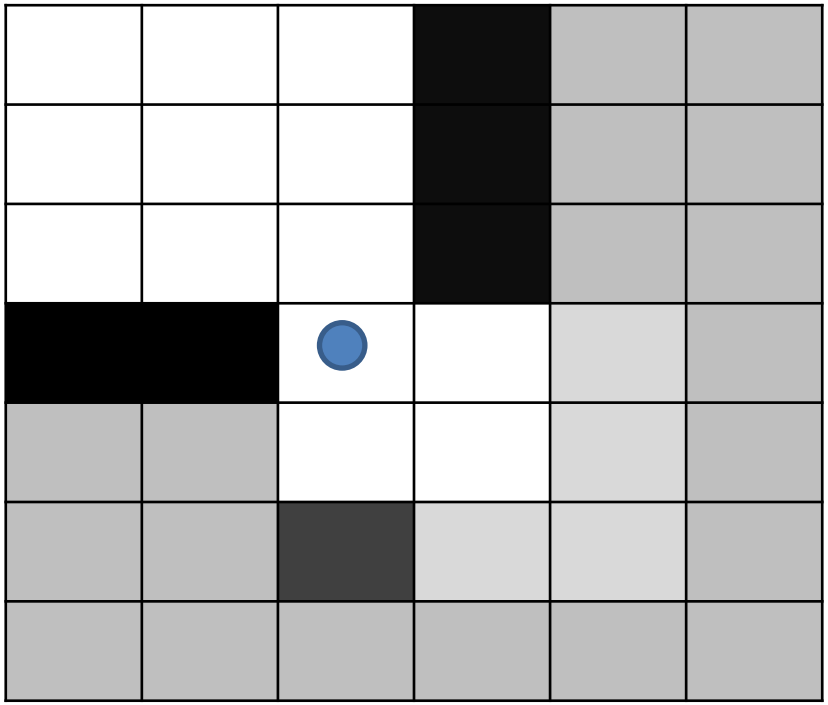
Mapping, Occupancy Grid



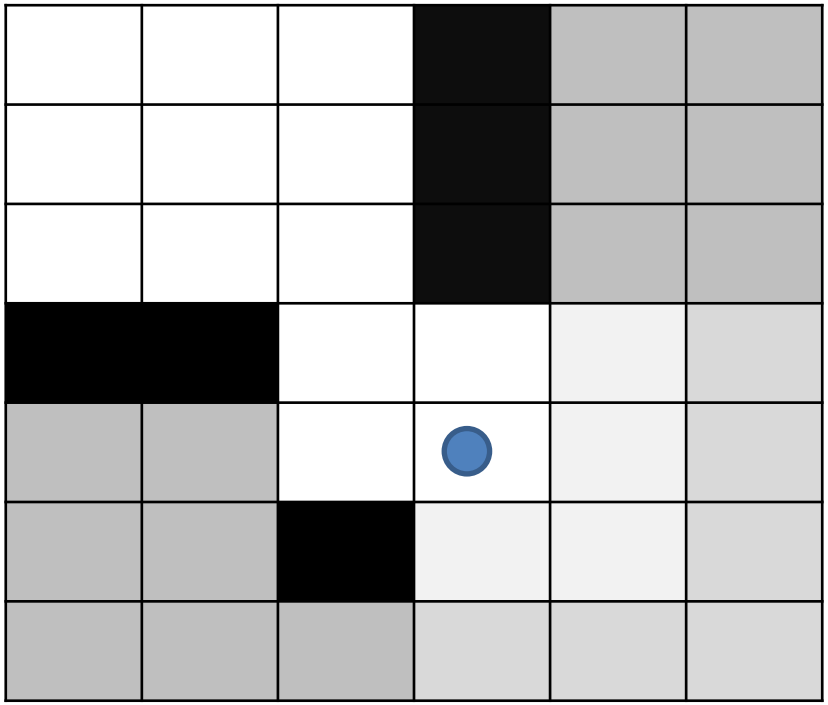
Mapping, Occupancy Grid



Mapping, Occupancy Grid



Mapping, Occupancy Grid



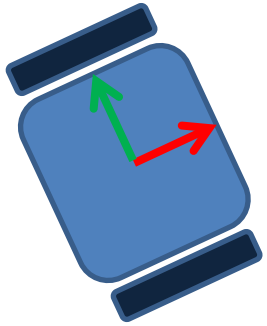
Mapping with Uncertain Robot Location

- Occupancy Grid
  - We need to consider how sure we are about our location and only update grid cells accordingly.
- Simultaneous Localization and Mapping



Simultaneous Localization and Mapping (SLAM)

- What if we do not have a map – but would like to create one?



?

Simultaneous Localization and Mapping (SLAM)

- What if we do not have a map – but would like to create one?
  - We can include the landmarks in our state vector.

$$x = [x_r, y_r, x_{l1}, y_{l1}, x_{l2}, y_{l2}, \cdots, x_{ln}, y_{ln}]$$

- The System model only updates x_r, y_r since the landmarks are not expected to change.
- The observation model as before but using the landmark positions *x*, instead of the mapping.



Simultaneous Localization and Mapping (SLAM)

- Sounds simple, but what are the challenges?



Simultaneous Localization and Mapping (SLAM)

- Sounds simple, but what are the challenges?
  - Dynamic size of the state
  - Initialization of covariance matrices
  - **The algorithm does not scale well to high dimensional state spaces**



Simultaneous Localization and Mapping (SLAM)

- Good algorithms exists, but unfortunately outside the scope of this course…
  - Requirements:
    - Know which problem to solve
    - Know the challenges and overall idea

- However, …
  - we will dive a bit deeper into this particular topic in two weeks time in a hands-on approach



Probabilistic Modeling beyond a Kalman Filter

- Do we always need to use a Kalman Filter?



Probabilistic Modeling beyond a Kalman Filter

- Notation
  - P(x(k)|u(0:k-1), y(1:k-1)): Probability of x(k) given inputs u(0) to u(k-1) and observations y(1) to y(k-1).
  - X: Space of all states x



Probabilistic Modeling beyond a Kalman Filter

- Prediction (Prior)

$$P(x(k)|u(0:k-1), y(1:k-1)) = \sum_{x(k-1)\epsilon X}(P(x(k)|u(k-1), x(k-1))P(x(k-1)|u(0:k-2), y(1:k-1)))$$

- Update (Posterior)

$$P(x(k)|u(0:k-1), y(1:k))$$

$$= \frac{P(y(k)|x(k))\ \ P(x(k)|u(0:k-1), y(1:k-1))}{\sum_{x(k)\epsilon X} P(y(k)|x(k))\ P(x(k)|u(0:k-1), y(1:k-1))}$$

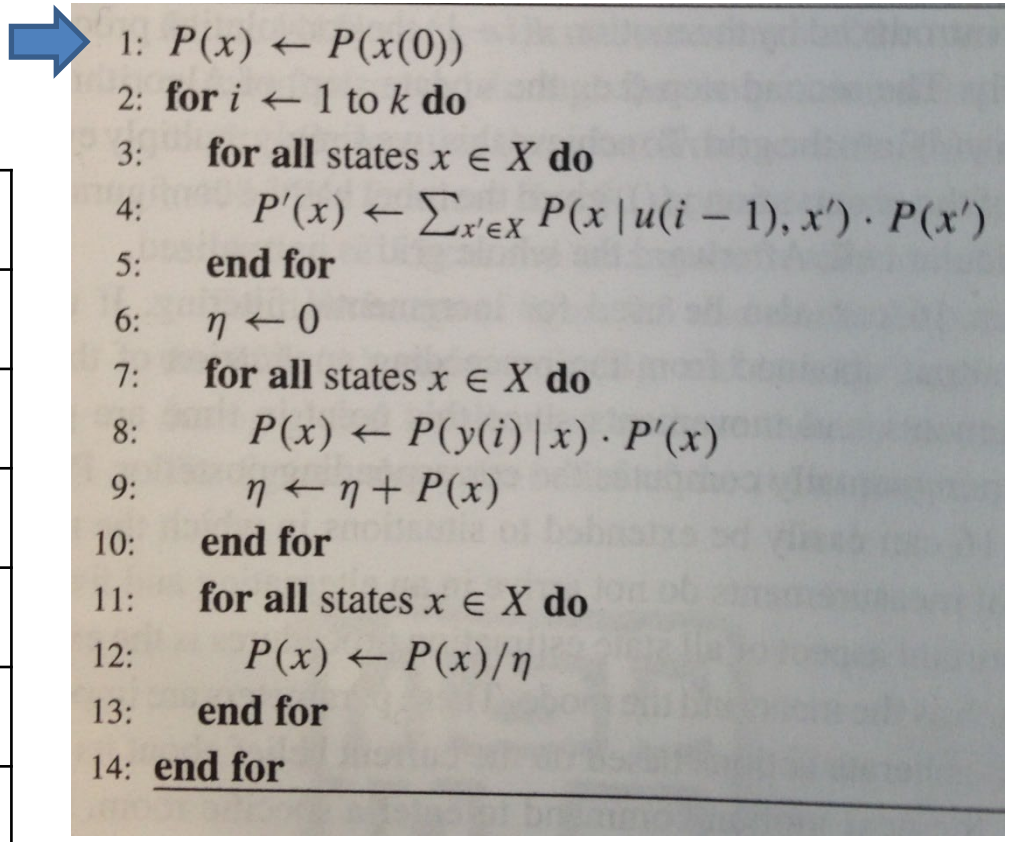
This is the general problem, not an actual implementation. The Kalman filter is so, though with certain assumptions and limitations.



Discrete Representation

Initial Distribution

	0.01	0.1	0.01		
	0.01	0.7	0.05		
	0.01	0.01	0.01		



1: $P(x) \leftarrow P(x(0))$
2: **for** $i \leftarrow 1$ to k **do**
3: **for all** states $x \in X$ **do**
4: $P'(x) \leftarrow \sum_{x' \in X} P(x \mid u(i-1), x') \cdot P(x')$
5: **end for**
6: $\eta \leftarrow 0$
7: **for all** states $x \in X$ **do**
8: $P(x) \leftarrow P(y(i) \mid x) \cdot P'(x)$
9: $\eta \leftarrow \eta + P(x)$
10: **end for**
11: **for all** states $x \in X$ **do**
12: $P(x) \leftarrow P(x)/\eta$
13: **end for**
14: **end for**

Discrete Representation

Distribution after prediction

		0.01	0.1	0.1	
		0.01	0.7	0.04	0.01
		0.01	0.01	0.01	



1: $P(x) \leftarrow P(x(0))$
2: **for** $i \leftarrow 1$ to k **do**
3: **for all** states $x \in X$ **do**
4: $P'(x) \leftarrow \sum_{x' \in X} P(x \mid u(i-1), x') \cdot P(x')$
5: **end for**
6: $\eta \leftarrow 0$
7: **for all** states $x \in X$ **do**
8: $P(x) \leftarrow P(y(i) \mid x) \cdot P'(x)$
9: $\eta \leftarrow \eta + P(x)$
10: **end for**
11: **for all** states $x \in X$ **do**
12: $P(x) \leftarrow P(x)/\eta$
13: **end for**
14: **end for**



Discrete Representation

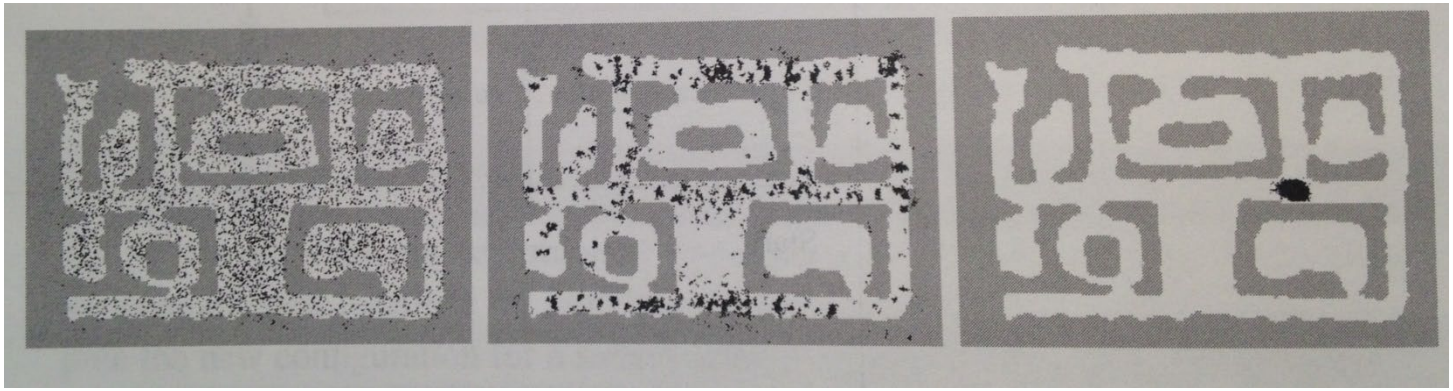
Distribution after update

			0.1	0.1	
		0.01	0.6	0.17	0.01
			0.01	0.01	

1: $P(x) \leftarrow P(x(0))$
2: **for** $i \leftarrow 1$ to k **do**
3: **for all** states $x \in X$ **do**
4: $P'(x) \leftarrow \sum_{x' \in X} P(x \mid u(i-1), x') \cdot P(x')$
5: **end for**
6: $\eta \leftarrow 0$
7: **for all** states $x \in X$ **do**
8: $P(x) \leftarrow P(y(i) \mid x) \cdot P'(x)$
9: $\eta \leftarrow \eta + P(x)$
10: **end for**
11: **for all** states $x \in X$ **do**
12: $P(x) \leftarrow P(x)/\eta$
13: **end for**
14: **end for**

Particle Filter

- Use a set of particles to represent the distribution



- Predict for each particle
- Update to see how likely a particle is
- Resample to keep the most important samples

Particle Filter

Algorithm 17 Probabilistic localization using a particle filter

Input: Sequence of measurements $y(1:k)$ and movements $u(0:k-1)$ and set $\mathcal{M}$ of N samples (x_j, ω_j) corresponding to the initial belief $P(x)$

Output: A posterior $P(x(k) \mid u(0:k-1), y(1:k))$ about the configuration of the robot at time step k represented by $\mathcal{M}$.

1: **for** $i \leftarrow 1$ to k **do**
2: **for** $j \leftarrow 1$ to N **do**
3: compute a new state x by sampling according to $P(x \mid u(i-1), x_j)$.
4: $x_j \leftarrow x$
5: **end for**
6: $\eta \leftarrow 0$
7: **for** $j \leftarrow 1$ to N **do**
8: $w_j = P(y(i) \mid x_j)$
9: $\eta = \eta + w_j$
10: **end for**
11: **for** $j \leftarrow 1$ to N **do**
12: $w_j = \eta^{-1} \cdot w_j$
13: **end for**
14: $\mathcal{M} = resample(\mathcal{M})$
15: **end for**



Particle Filter

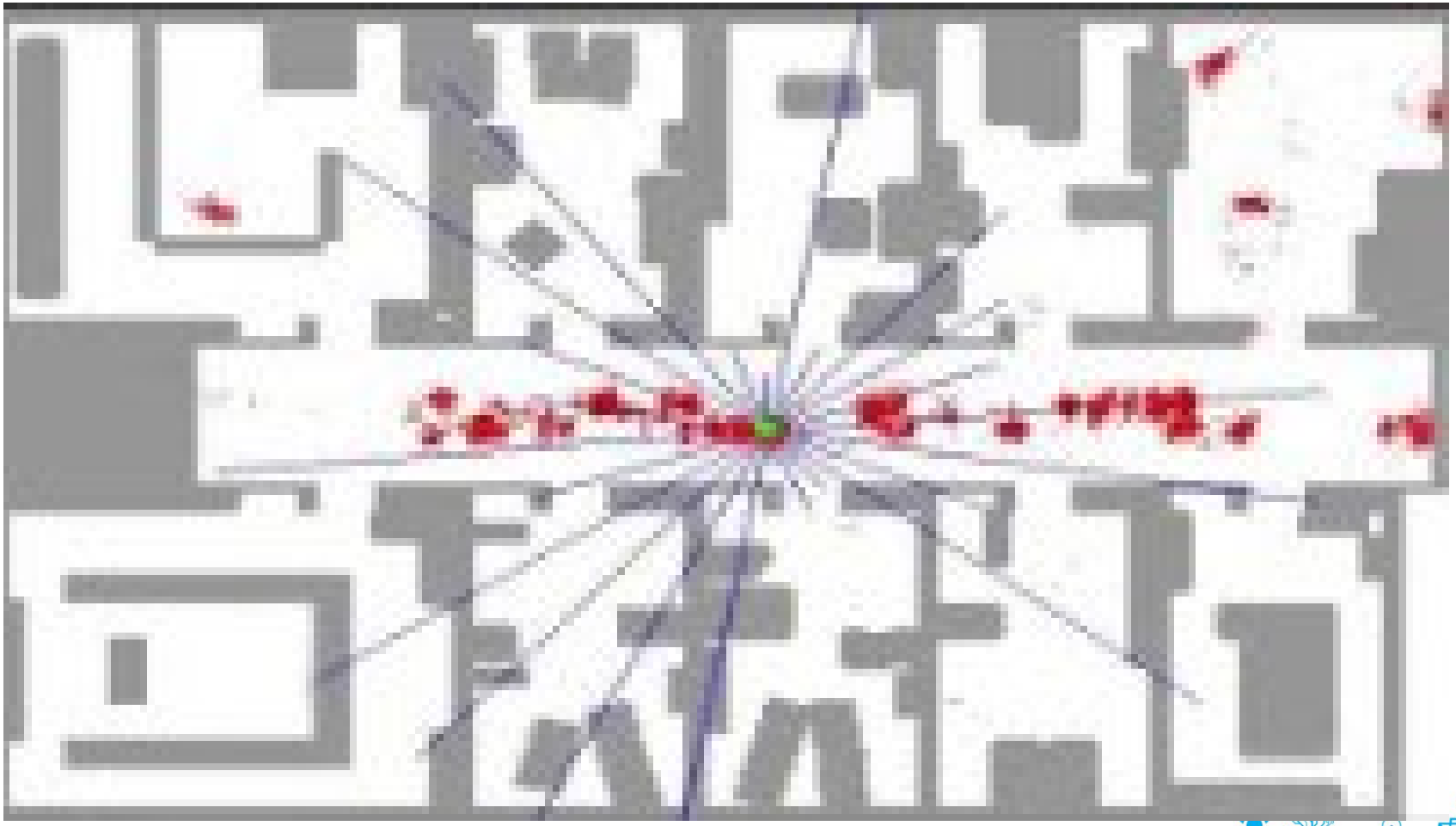
Algorithm 18 The procedure *resample*($\mathcal{M}$)

Input: Set $\mathcal{M}$ of N samples

Output: Set $\mathcal{M}'$ of N samples obtained by importance resampling from $\mathcal{M}$

1: $\mathcal{M}' \leftarrow \emptyset$
2: $\Delta \leftarrow rand((0; N^{-1}])$
3: $c \leftarrow \omega_0$
4: $i \leftarrow 0$
5: **for** $j \leftarrow 0$ to $N - 1$ **do**
6: $\quad u \leftarrow \Delta + j \cdot N^{-1}$
7: $\quad$ **while** $u > c$ **do**
8: $\quad\quad i \leftarrow i + 1$
9: $\quad\quad c \leftarrow c + \omega_i$
10: $\quad$ **end while**
11: $\quad \mathcal{M}' \leftarrow \mathcal{M}' \cup \{(x_i, N^{-1})\}$
12: **end for**





- Summary
  - Kalman Filter
  - Extended Kalman Filter
  - Particle Filter



Programming Exercise 8

Consider a simple mobile robot

- Implement a Particle Filter (PF) for localization.
- For development, assume a sand-box case
  - Robot moving in a straight line, no errors
  - The observer reports the position (e.g., mimicking GNSS with some isotropic error)
  - Focus on the structure of the PF

- Adopt the case, e.g.:
  - Change the observer to report range and bearing to some fixed (virtual) features
  - Consider realistic limitations: limited range, noise, uncertainty connected to motion (errors)
  - You may want to consider introducing rarer errors, mimicking sudden slip etc.

- Evaluate the solution
  - Define an evaluation strategy (e.g., assess the impact of the number of particles, the ability to recover from errors, the effect of parameters, e.g., bandwidth of distributions used for the PF

